Buprenorphine for the Management of 7-Hydroxymitragynine (7-OH) Use: A Retrospective Case Series

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Background: Kratom (*Mitragyna speciosa*) is increasingly used in the United States as a dietary supplement for pain, mood regulation, and opioid withdrawal. Its alkaloids, mitragynine and 7-hydroxymitragynine (7-OH), demonstrate μ-opioid receptor activity, with 7-OH exhibiting substantially greater opioid-like potency than mitragynine. There are concerns from the US Food and Drug Administration (FDA) regarding dependence and adverse effects. Purified 7-OH products may bypass metabolic pathways and deliver disproportionately strong opioid effects compared with natural kratom preparations, increasing the risk of dependence. Despite these concerns, 7-OH remains federally unscheduled, though listed as a "Drug of Concern" by the DEA. Although there is no standardized clinical guidance, buprenorphine has been used to manage kratom and 7-OH withdrawal with varying initiation strategies.

Objectives: To describe the clinical course and outcomes of buprenorphine initiation among patients with problematic 7-OH use, focusing on initiation timing, dosing strategies, and symptom monitoring.

Methods: Retrospective chart review of patients treated at a low-barrier telehealth addiction medicine clinic between April and October 2025.

Results: Nine patients with purified 7-OH product use (as opposed to kratom use) met the inclusion criteria. The mean age was 33.5 years; 7 patients were identified as male. Low-dose initiation was used in 6 cases and standard initiation in 3 cases. Successful initiation and stabilization occurred in 88.9%, with no precipitated withdrawal or adverse events. Symptom improvement was reported in 8 of the total cases at a median 6-week follow-up.

Conclusions: Buprenorphine initiation for problematic 7-OH use was feasible, well tolerated, and associated with symptom improvement, supporting development of pragmatic clinical approaches for this emerging substance use disorder.

Key Words: kratom, 7-hydroxymitragynine (7-OH), buprenorphine, psychoactive substance.

(*J Addict Med* 2026;00:000–000)

Kratom (*Mitragyna speciosa*) is a plant-derived substance with increasing use in the United States, particularly for self-management of chronic pain, mood symptoms, and withdrawal from other substances. Its primary alkaloids, mitragynine and 7-hydroxymitragynine (7-OH), are partial agonists at the μ-opioid receptor, with 7-OH being a higher-potency partial agonist, though human clinical potency and risk remain incompletely characterized, prompting regulatory scrutiny in the face of risk of dependence. In July 2025, the FDA issued several warning letters to companies for illegally marketing 7-OH. Naturally occurring levels of 7-OH are found in minimal amounts (~0.01%) within kratom leaves and further undergo CYP-mediated conversion, leaving negligible remaining amounts.[1] Unlike naturally occurring 7-OH, purified and concentrated 7-OH products present a heightened risk of dependence and opioid-like adverse effects, which have been predominantly extrapolated from animal and mechanistic studies.[2] These purified preparations may bypass the slower metabolic conversion of mitragynine to 7-OH, thereby delivering disproportionately strong opioid-like activity, though the clinical impact of partial agonism on the μ-opioid receptor has not been well characterized.[1,3–8] 7-OH is ~4–7 times more potent than morphine in mouse antinociception models and has been shown to have minimal abuse liability, reduce opioid tolerance when administered before morphine, and have a lack of β-arrestin recruitment.[8–12] Such formulations raise safety concerns

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Received for publication February 7, 2026; accepted May 8, 2026.

The authors report no conflicts of interest.

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Supplemental Digital Content is available for this article. Direct URL citations are provided in the HTML and PDF versions of this article on the journal's website, www.journaladdictionmedicine.com.

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ISSN: 1932-0620/26/0000-0000

DOI: 10.1097/ADM.0000000000001723

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given their marketing as dietary supplements or "natural" remedies without standardized dosing or regulatory oversight. While kratom and 7-OH remain unscheduled at the federal level, their pharmacology raises significant concerns for both dependence and subsequent withdrawal. Clinical presentations suggest that individuals who consistently use 7-OH develop tolerance, escalating use, and withdrawal syndromes when attempting to stop, thereby meeting several standard DSM-5 criteria for a substance use disorder. Despite these risks, kratom and 7-OH continue to be sold as herbal supplements (often in low-barrier settings such as corner stores and gas stations under the guise of natural remedies for pain, mood, and energy), with limited regulation and clinical guidance for clinicians managing affected patients.

Emerging clinical reports describe problematic 7-OH use consistent with substance use disorder features as an underrecognized, yet growing phenomenon.[1,10,13] Unlike traditional opioid use disorders, 7-OH withdrawal may present with a somewhat unique symptom profile, including pronounced anxiety, irritability, insomnia, and restlessness, in addition to the classic autonomic and gastrointestinal features otherwise seen with kratom and opioid withdrawal.[14] Patients may use 7-OH or kratom specifically to avoid withdrawal from prescription opioids or fentanyl, yet ultimately develop physiological dependence on the substance itself. Increasingly, patients are presenting to primary care, emergency, and addiction medicine settings seeking help with discontinuation due to physical and financial dependence on the substance, though clinicians face uncertainty about effective and safe treatment strategies.[15]

Buprenorphine, a well-known partial μ-opioid receptor agonist, is the standard of care for opioid use disorder and has been used off-label (alongside methadone and naltrexone) in case reports and series to manage kratom dependence, though there is no dedicated research for the management of 7-OH use.[10,16] The absence of a structured, reproducible protocol limits generalizability and leaves uncertainty about how to approach buprenorphine initiation in patients using 7-OH. Furthermore, the natural history and outcomes of buprenorphine treatment for kratom use disorder, especially in outpatient settings, remain poorly described.

Given the rising prevalence of patients using 7-OH and the lack of clinically pragmatic approaches applicable in real-world addiction treatment settings amidst a growing clinical need, a structured examination of buprenorphine initiation in this population is of the utmost importance. This case series provides observational descriptions of patient presentations, initiation strategies, and short-term outcomes. By capturing both clinical course and patient-reported experiences, this case series aims to generate foundational knowledge that can guide safe and effective management of 7-OH withdrawal in clinical practice. These findings will serve as preliminary data to inform the development of standardized initiation protocols and ultimately support larger prospective studies.

METHODS

Study Design and Setting

This study was a retrospective, observational case series conducted within the Harm Reduction & BRidges to Care (HRBR) outpatient addiction medicine clinic. The case series was designed to descriptively characterize clinical presentations, buprenorphine initiation approaches, and stabilization trajectories among patients presenting with use of purified 7-OH. This case series was conducted and reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for observational studies. This study was reviewed by the Oregon Health and Science University Institutional Review Board (Study #29198) and was determined not to constitute human subjects research and therefore was exempt from further IRB review. As the study involved retrospective review of existing clinical records with no direct patient contact, intervention, or collection of identifiable information, the requirement for informed patient consent was waived.

Case Identification

Cases were identified through review of the electronic health record (EHR) for patients receiving care at the HRBR clinic who reported current use of 7-OH and underwent buprenorphine initiation during routine clinical care. Identification was based on clinician documentation of 7-OH use in intake notes, follow-up encounters, or substance use histories. Inclusion and exclusion criteria are further detailed in Appendix A, Supplemental Digital Content 1, http://links.lww.com/JAM/A815

Data Sources and Abstraction

Data were extracted from the EHR and entered into a secure database. Each case was assigned a unique study identifier, and all data were de-identified before analysis.

Chart abstraction was performed using a structured abstraction form developed to capture demographic, clinical, and treatment-related variables. Extracted variables included patient demographics, reported 7-OH use patterns, buprenorphine initiation strategy, total daily buprenorphine dose during initiation as well as stabilization, adjunct medications, documented clinical outcomes, and incidence of concurrent or history of other substance use disorders.

A qualitative review of clinician notes was conducted to capture patient-reported symptoms, withdrawal experiences, tolerability of initiation, and functional impacts. These qualitative data were summarized narratively to illustrate representative clinical courses and contextualize quantitative findings.

Definitions

- Exposure (7-OH use): Defined as patient self-report of current use of purified 7-hydroxymitragynine products at the time of presentation. Reported daily use was documented either in milligrams or, when unavailable, as estimated daily expenditure.

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- Concurrent substance use disorders (SUDs): Defined by clinician-documented diagnoses in the medical record based on DSM-5 criteria or treatment history.
- Low-dose buprenorphine initiation: Defined as initiation of buprenorphine using incremental, subtherapeutic doses (eg, starting at ~0.5–1 mg total daily dose, with stepwise increases to 2–4 mg over the first 1–2 d) while a full opioid agonist is continued (over days 1–2), followed by gradual dose escalation over multiple days (typically reaching ≥8–16 mg/d) to achieve a therapeutic maintenance dose.
- Standard buprenorphine initiation: Defined as initiation of buprenorphine after the onset of moderate opioid withdrawal (COWS score ≥8–12 and often ≥24 hours following last opioid use), with administration of typical starting doses (eg, 2–4 mg initial dose, with additional 2–8 mg on day 1 as needed, up to ~8–12 mg total on day 1) and subsequent titration to a therapeutic maintenance dose (generally 8–24+ mg/d).
- High-dose buprenorphine initiation: Defined as a large initial dose (≥16 mg as the first dose) in patients with fentanyl exposure and severe withdrawal or >12 mg total buprenorphine on the first day.[17,18]
- Initiation success: Defined as completion of buprenorphine initiation (3 or more days on stable dose) without discontinuation due to intolerable symptoms or precipitated withdrawal.
- Stabilization: Defined as clinician-documented suppression of withdrawal symptoms and cravings with a stable total daily buprenorphine dose for three or more days.
- Symptom improvement: Defined as clinician-documented reduction in patient-reported withdrawal symptoms, cravings, or associated neuropsychiatric symptoms (eg, anxiety, insomnia, irritability) following buprenorphine initiation, as noted in follow-up clinical documentation.
- Precipitated withdrawal: Defined as a rapid and clinically significant worsening of opioid withdrawal symptoms within hours of buprenorphine administration, consistent with published operational criteria. Operationalized as an acute increase in withdrawal severity (eg, ≥6-point increase in COWS score or clear clinical deterioration) temporally associated with buprenorphine initiation and documented in the shared health record, without an alternative explanation.[19,20]
- Treatment engagement: Defined as receipt of MOUD during the 30 days following initiation.[21]

Urine drug screening was not uniformly obtained and, therefore, was not used as a required criterion for confirmation of exposure or concurrent substance use, particularly in the absence of routine laboratory testing for kratom and 7-OH. Ascertainment of precipitated withdrawal and adverse events relied on subjective history as well as review of outpatient clinic documentation and available emergency department or inpatient records within the same health system.

Outcomes

Clinical outcomes of interest included initiation approach, whether initiation was completed, achievement of stabilization, maximum buprenorphine dose required, presence or absence of precipitated withdrawal, symptom trajectories during initiation, and engagement in care at follow-up.

Data Analysis

Given the exploratory nature and small sample size of this retrospective case series, analyses were descriptive. Patient demographic and clinical characteristics were summarized using measures of central tendency and range. Continuous variables (eg, age, estimated daily 7-OH use, duration of use) were reported as a mean, median, or range, depending on the variable itself. Categorical variables (eg, sex, presence of concurrent SUDs, initiation strategy, stabilization status) were summarized using frequencies and proportions.

No inferential statistical analyses or hypothesis testing were performed due to the case series design, limited cohort size, and heterogeneity of presentations. Missing data were uncommon and occurred primarily when 7-OH exposure was documented in monetarily rather than milligram units. Analyses were conducted to descriptively characterize observed clinical practices and generate hypotheses for future studies.

RESULTS

A total of 9 clinical cases involving buprenorphine treatment for 7-OH use were identified and included in this case series (Table 1).

The mean age of patients was 33.5 years (range: 20–46). Most patients identified as male (n = 7, 77.7%), with one female patient and one who identified as nonbinary. The cohort was predominantly White (n = 7, 77.7%), with 2 patients reporting other racial or ethnic identities. Concurrent substance use disorders were present in 66.6% (n = 6) of cases.

Reported 7-OH use varied substantially. When reported in milligrams (n = 7), the median daily dose was 150 mg/d; when recorded as monetary expenditure (n = 2), the median expenditure was $33.14/d. The mean duration of 7-OH use before presentation was 11.4 months.

Buprenorphine initiation strategies included both standard and low-dose approaches, with 6 patients who underwent low-dose initiation and 3 patients who underwent standard initiation. Successful initiation was achieved in 88.9% of cases, and all 9 patients remained engaged in treatment with buprenorphine. Symptom improvement was reported in 8 of the total cases. No precipitated withdrawal was observed. The mean standard initial total daily dose (TDD) was 4 mg (median 2 mg), while the mean low-dose initial TDD was 4.25 mg (median 1.25 mg). Most patients (n = 8) required moderate to high stabilization doses (≥16 mg TDD), with several cases necessitating 24–32 mg/d to adequately suppress cravings and disrupt ongoing 7-OH use; 1 patient required a buprenorphine dose ≤16 mg TDD, which was predominantly driven by patient preference. Stabilization was achieved in 8 cases. One patient declined dose esca-

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TABLE 1. Clinical Cases of Buprenorphine for 7-OH Use

Patient Demographics	Daily 7-OH Dose (mg or $/d If Not Provided in mg)	Duration of 7-OH Use (mo)	Buprenorphine Stabilization Dose (TDD – mg) and Formulation	Initiation Method	Supportive Meds Used	Stabilization Achieved?	Important Case Considerations
Clinic patient #1 39 y/o M with h/o OUD, GAD, MDD, and ADHD	98–128	13	16, SL bupe-nal tablets	Low dose	Tizanidine	Yes	The initiation dose was too low and required 12 mg ~3 d following initial dosing. Stabilized at 16 mg
Clinic patient #2 28 y/o M with OCD, GAD	100–150	4	16, SL bupe-nal tablets	Standard	None	Yes	Was on 12 mg through HRBR clinic and eventually increased to 16 mg TDD through PCP follow-up
Clinic patient #3 25 y/o M with h/o OUD, PTSD, bipolar 2, ADHD, GAD	$30–40/d	1	24, SL bupe mono tablets	Low dose	Clonidine, tizanidine, ondansetron	Yes	Cravings on 16 mg TDD prompting increase to 24 mg
Clinic patient #4 46 y/o F with GAD and insomnia	300	12	16, SL bupe-nal tablets	Low dose	Trazodone, clonidine, ondansetron, hydroxyzine, tizanidine, loperamide, ibuprofen	Yes	Initially opted for detox from 7-OH, though later requested MAT
Clinic patient #5 24 y/o nonbinary patient with ADHD, GAD, chronic pain, gender dysphoria fibromyalgia, MDD, and unspecified psychosis	$100/d	5	24, SL bupe mono tablets	Standard	Ondansetron, clonidine, buspirone, ibuprofen	Yes	Initially had buprenorphine initiation while in active withdrawal before planned surgery, max dose 28 mg TDD though stabilized on 24 mg TDD
Clinic Patient #6 37 y/o M with GAD, MDD, OUD (in remission), and AUD (in remission)	400	48	28, SL bupe mono tablets	Low dose	Clonidine, tizanidine, gabapentin, hydroxyzine	Yes	Initial buprenorphine initiation was unsuccessful (3-d low dose). Reached 22 mg TDD but unable to disrupt use), went to medically managed withdrawal (detox) and was started on buprenorphine initial dose of 40 mg on day 1, then stabilized at 24–28 mg TDD. Later tapered with the intention to give LAI as taper off
Clinic patient #7 20 y/o M with GAD, MDD, insomnia, ADHD, and chronic pain	270	11	32, SL bupe mono tablets	Low dose	Trazodone, tizanidine	Yes	Difficulty decreasing 7-OH use despite high buprenorphine doses
Clinic patient #8 37 y/o M with chronic pain, OUD (in remission) and insomnia	150	3	4, SL bupe-nal tablets	Standard	Tizanidine, clonidine, hydroxyzine	No	Decreased 7-OH use on MAT, though did not want to increase higher than 4 mg TDD.
Clinic patient #9 46 y/o M with chronic pain, methamphetamine use disorder (in remission), insomnia, and MDD	300–400	6	28, SL bupe-nal tablets	Low-dose	Clonidine, tizanidine, ondansetron, hydroxyzine, loperamide, ibuprofen, trazodone, mirtazapine, bupropion	Yes	Significant anhedonia and insomnia with disruption of 7-OH requiring both mirtazapine temporarily and later bupropion

Bupe-nal, buprenorphine–naloxone; bupe mono, buprenorphine monoproduct; HRBR, Harm Reduction & BRidges to Care; LAI, long-acting injectable; MAT, medication-assisted treatment; SL, sublingual; TDD, total daily dose.

Clinical characteristics, 7-OH use patterns, and buprenorphine initiation and stabilization details among patients included in this observational case series.

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lation beyond 4 mg/d and did not achieve full stabilization despite reduced 7-OH use.

Adjunct medications, including clonidine, tizanidine, antiemetics, and antidepressants, were commonly used to manage withdrawal symptoms, insomnia, anxiety, or mood disturbance during initiation and stabilization. Several cases highlighted the need for dose adjustments, prolonged initiation periods, or higher-than-typical buprenorphine doses, reflecting the clinical complexity of treating 7-OH dependence.

Figure 1 illustrates the clinical course of buprenorphine initiation among patients with 7-hydroxymitragynine (7-OH) use (n = 9), demonstrating 2 distinct initiation pathways and their convergence on shared clinical outcomes while highlighting the absence of precipitated withdrawal in all cases. The majority of patients who underwent low-dose initiation achieved gradual symptom improvement and stabilization over the first 2 weeks of treatment. There was a modest positive relationship between self-reported 7-OH intake and the buprenorphine dose required for stabilization. Patients reporting higher estimated daily 7-OH consumption tended to require higher buprenorphine doses to achieve symptom control, though stabilization was achieved across all dosing ranges.

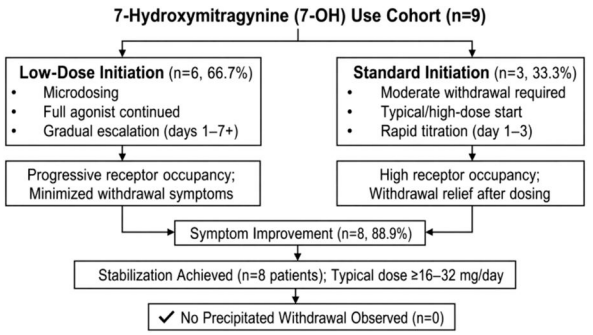


FIGURE 1. Flowchart illustrating initiation protocols, symptom improvement, stabilization, and absence of withdrawal symptoms in the 7-hydroxymitraginine cohort.

REPRESENTATIVE CASE NARRATIVES

Case 1

A 46-year-old male presented to Harm Reduction & BRidges to Care clinic (HRBR) using 300–400 mg/d of 7-OH in the preceding 6 months, ultimately spending $1500–2000/month. He made attempts to adjust his pain regimen when his chronic back pain worsened. He was given advice from the store owner that 7-OH would be more effective for pain control than the standard kratom he had been taking for the past 3 years. With provider guidance, the patient attempted low-dose buprenorphine initiation over a 7-day period, with an initial TDD of 0.5 mg and a max dose of 28 mg daily. The week following his disruption of 7-OH use was compounded by fatigue, anhedonia, and insomnia, making it challenging for him to return to work. HRBR providers trialed trazodone and clonidine for insomnia, which were ineffective. A short course (3 d) of mirtazapine 15 mg was then utilized with adjunct prescription bupropion 150 mg XL for new onset depression/fatigue. He later stabilized on a 28 mg total daily dose (TDD) with no return to 7-OH or kratom use. Following this, the patient returned to the corner store where he would typically buy 7-OH to advise store personnel to remove this dangerous substance from the shelves.

Case 2

A 37-year-old male presented to HRBR using ~400 mg/d 7-OH daily at the time of initial presentation. The patient attempted low-dose initiation with an initial TDD of 2 mg and eventually increased his buprenorphine to 22 mg TDD ~1 month following initial initiation, though he was unsuccessful at disrupting 7-OH use. He then presented to an inpatient medically managed withdrawal (detox) center, during which time he was started on buprenorphine via a high-dose approach. The patient reported that on the first day of buprenorphine initiation, he took 40 mg of buprenorphine TDD. On day 2, he took 32 mg TDD. Following this, he met with an HRBR provider at which time he noted he had reduced his dose to 24 mg TDD before discharge from the medically managed withdrawal center. He was stabilized at 24–28 mg buprenorphine TDD and remained engaged in treatment. He then successfully tapered his buprenorphine to 16 mg buprenorphine TDD over 3 months. His long-term goal is to taper off buprenorphine via a one-time dose of long-acting injectable (LAI) buprenorphine.

DISCUSSION

In this retrospective, observational case series, we report one of the first case series with clinical descriptions of off-label buprenorphine initiation for patients using 7-OH, an increasingly encountered kratom-derived alkaloid in clinical settings. As a convenience-sampled case series, these findings are descriptive and hypothesis-generating in nature and are not intended to demonstrate treatment efficacy, establish causality, or define standards of care. Clinicians in our case series utilized shared decision-making with patients to offer buprenorphine for off-label use as one component of an individualized treatment plan.

Across cases, patients exhibited features consistent with physiological opioid-like dependence and correspondingly, buprenorphine initiation appeared well tolerated in this cohort. However, this observation should be interpreted cautiously and reflects clinical experience rather than comparative effectiveness. Both low-dose and standard initiation approaches were used, potentially reflecting clinical uncertainty in the absence of established guidance for managing dependence on high-potency kratom-derived products.

Several patients required higher buprenorphine doses than anticipated (≥ 16 mg TDD) to achieve stabilization, which is higher than traditionally reported for lower-potency opioid exposures. Higher maintenance

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doses were often required compared with typical opioid use disorder treatment, suggesting that higher receptor occupancy may be necessary in this population. Following successful high-dose initiation, rapid down-titration (eg, from ≥ 32 mg to 24–28 mg total daily dose) appeared feasible in select cases without clinical destabilization once 7-OH use was fully disrupted, particularly when weighing the risks and benefits of continued buprenorphine treatment in an evidence-limited context. Many patients required doses like those used for fentanyl-associated opioid use disorder; these findings should not be interpreted as implying equivalence in pharmacologic potency or receptor dynamics but rather reflect observed clinical decision-making.

Importantly, disruption of high-dose 7-OH use was frequently associated with persistent neuropsychiatric symptoms (eg, anhedonia, fatigue, profound anxiety, and insomnia), which in some cases extended beyond the acute withdrawal period. Standard supportive medications, such as clonidine or trazodone, were not always sufficient to manage these symptoms, highlighting the need for anticipatory guidance and comprehensive symptom management strategies during early recovery. Despite these challenges, successful stabilization with buprenorphine was associated with sustained abstinence from 7-OH in many of our patients, including those with prolonged and high-dose use.

Table 2 is presented as a descriptive synthesis of observed dosing patterns, not as proposed treatment guidelines. Its purpose is to contextualize the range of buprenorphine doses used in this case series according to reported 7-OH exposure, facilitating clinical understanding and hypothesis generation. This approach parallels prior descriptive work in emerging substance use contexts, including the case series by Weiss and Douglas,[15] which similarly summarized clinical dosing experiences without asserting efficacy or prescriptive recommendations. Consistent with that precedent, Table 2 is intended to inform future systematic investigation rather than guide clinical practice. The case series by Weiss and Douglas included 9 patients using kratom products and found a linear relationship between kratom use and buprenorphine initiation dose, similar to the linear relationship that was evident in our cohort of patients using 7-OH. However, we must also acknowledge that the potency of 7-OH products varies significantly in a largely unregulated market and may contain mitragynine or mitragynine pseudoindoxyl, a more potent 7-OH metabolite. As such, this variability precludes specific buprenorphine dose recommendations based on this small sample.[22]

Our findings add to a small but growing literature suggesting that buprenorphine is a rational and effective treatment for kratom-derived alkaloid dependence. Given variability in pharmacokinetics between traditional kratom and purified 7-OH products, this focused analysis provides preliminary evidence that low-dose initiation may be particularly useful where potency and metabolic dynamics raise concerns for precipitated withdrawal.

TABLE 2. Observed Buprenorphine Dosing Patterns Based on Daily 7-OH Use

Estimated Daily 7-OH Use (mg/d)	Estimated Kratom Doses (per Weiss et al),[15] g kratom/d	Rationale	Proposed Buprenorphine Dose Range (mg/d)	Notes
< 50	< 20	Lower physiological tolerance, mild withdrawal risk	2–4	Consider standard initiation if already in moderate withdrawal
50–150		Moderate dependence, risk of dysphoria/anxiety with cessation	4–8	Low-dose initiation is optional if unable to tolerate abstinence/withdrawal symptoms
150–300	> 40	Significant opioid-like tolerance and withdrawal	8–16	Low-dose initiation preferred; consider split or incremental dosing
> 300		High-tolerance pattern similar to high-dose opioid use	12–24+	Low-dose initiation strongly recommended; higher doses may be required for stabilization

Observed buprenorphine total daily dose ranges associated with clinical stabilization in patients reporting use of purified 7-OH in this observational case series, with comparison to estimated kratom doses as demonstrated in the study by Weiss and colleagues.

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Limitations

This study has several important limitations. Small sample size, retrospective design, single-clinic setting, occasional incomplete dosing documentation, and reliance on patient self-report of 7-OH consumption not verified by toxicological testing limit generalizability. Selection bias is inherent in the harm reduction clinical context, as patients who engage in care may be different from those who do not, and follow-up duration varies across cases. Further, because the sample consisted primarily of White male patients, in part representative of the geographic area at large, these observations may not fully capture patterns of 7-OH use or treatment experiences among other diverse populations.

The variability of 7-OH products available and the risk of mislabeling limit the standardization of the exposure. While efforts were made to apply consistent operational definitions for key outcomes (eg, initiation success, stabilization, and symptom improvement), outcomes were based on clinician documentation rather than standardized assessment tools, which may limit comparability across cases. As such, conclusions regarding comparative initiation strategies, optimal dosing, or treatment effectiveness cannot be drawn.

Future Directions

Despite these limitations, this case series provides early descriptive insight into the clinical complexity of managing 7-OH use and highlights the range of buprenorphine-based approaches currently employed in practice. As kratom-derived alkaloids, including 7-OH, continue to increase in concentration and availability, systematic prospective studies are needed to better characterize withdrawal syndromes, initiation tolerability, and longer-term outcomes. Until such data are available, descriptive case series may help inform clinical awareness and identify priorities for future research.

CONCLUSIONS

This observational case series is one of the first that describes clinical experiences with buprenorphine initiation among patients presenting with use of purified 7-OH as feasible, well tolerated, and associated with high rates of stabilization and symptom improvement in a real-world outpatient setting. Both low-dose and standard initiation approaches were successfully utilized without precipitated withdrawal or adverse events.

As the availability and potency of 7-OH products continue to increase, further prospective studies are needed to better characterize withdrawal syndromes, optimize initiation strategies, and define longer-term outcomes to guide evidence-based care.

REFERENCES

1. Todd DA, Kellogg JJ, Wallace ED, et al. Chemical composition and biological effects of kratom (Mitragyna speciosa): in vitro studies with implications for efficacy and drug interactions. *Sci Rep*. 2020;10(1):19158.
2. U.S. Food and Drug Administration. Products containing 7-OH can cause serious harm. Accessed September 11, 2025. https://www.fda.gov/consumers/consumer-updates/products-containing-7-oh-can-cause-serious-harm
3. Henningfield JE, Fant RV, Wang DW. The abuse potential of kratom according the 8 factors of the controlled substances act: implications for regulation and research. *Psychopharmacology (Berl)*. 2018;235(2):573–589.
4. Eggleston W, Stoppacher R, Suen K, et al. Kratom use and toxicities in the United States. *Pharmacotherapy*. 2019;39(7):775–777.
5. Cinosi E, Martinotti G, Simonato P, et al. Following "the roots" of kratom (*Mitragyna speciosa*): the evolution of an enhancer from a traditional use to increase work and productivity in Southeast Asia to a recreational psychoactive drug in western countries. *Biomed Res Int*. 2015;2015:968786.
6. U.S. Food and Drug Administration 2025. FDA takes steps to restrict 7-OH opioid products threatening American consumers. https://www.fda.gov/news-events/press-announcements/fda-takes-steps-restrict-7-oh-opioid-products-threatening-american-consumers
7. Holler JM, Vorce SP, McDonough-Bender PC, et al. A drug toxicity death involving propylhexedrine and mitragynine. *J Anal Toxicol*. 2011;35(1):54–59.
8. Wightman RS, Hu D. A case of 7-OH mitragynine use requiring inpatient medically managed withdrawal. *J Addict Med*. 2025. doi:10.1097/ADM.0000000000001558
9. Hill R, Kruegel AC, Javitch JA, et al. The respiratory depressant effects of mitragynine are limited by its conversion to 7-OH mitragynine. *Br J Pharmacol*. 2022;179(14):3875–3885.
10. Stanciu C, Ahmed S, Gnanasegaram S, et al. Kratom as an opioid alternative: harm, or harm reduction? A systematic review of literature. *Am J Drug Alcohol Abuse*. 2022;48(5):509–528.
11. Matsumoto K, Hatori Y, Murayama T, et al. Involvement of mu-opioid receptors in antinociception and inhibition of gastrointestinal transit induced by 7-hydroxymitragynine, isolated from Thai herbal medicine Mitragyna speciosa. *Eur J Pharmacol*. 2006;549(1-3):63–70.
12. Matsumoto K, Horie S, Ishikawa H, et al. Antinociceptive effect of 7-hydroxymitragynine in mice: discovery of an orally active opioid analgesic from the Thai medicinal herb Mitragyna speciosa. *Life Sci*. 2004;74(17):2143–2155.
13. Stanciu C, Ahmed S, Hybki B, et al. Pharmacotherapy for management of 'kratom use disorder': a systematic literature review with survey of experts. *WMJ*. 2021;120(1):54–61.
14. Buresh M. Treatment of kratom dependence with buprenorphine-naloxone maintenance. *J Addict Med*. 2018;12(6):481–483.
15. Weiss ST, Douglas HE. Treatment of kratom withdrawal and dependence with buprenorphine/naloxone: a case series and systematic literature review. *J Addict Med*. 2021;15(2):167–172.
16. Sherrick RC. Treatment of kratom use disorder with methadone in an opioid treatment program. *J Addict Med*. 2026. doi:10.1097/ADM.0000000000000166.
17. Parke B, Buchheit BM, Teeter-Balin E. High-dose buprenorphine initiation delivered in a low-barrier telemedicine clinic: a case series for fentanyl use treatment. *J Subst Use Addict Treat*. 2026;187:209968.
18. Wong S, Fabiano N, Webber D, et al. High-dose buprenorphine initiation: a scoping review. *J Addict Med*. 2024;18(4):349–359.
19. Dunn KE, Bird HE, Bergeria CL, et al. Operational definition of precipitated opioid withdrawal. *Front Psychiatry*. 2023;14:1141980.
20. D'Onofrio G, Hawk KF, Perrone J, et al. Incidence of precipitated withdrawal during a multisite emergency department-initiated buprenorphine clinical trial in the era of fentanyl. *JAMA Netw Open*. 2023;6(3):e236108.
21. McNeely J, Wang SS, Rostam Abadi Y, et al. Addiction consultation services for opioid use disorder treatment initiation and engagement: a randomized clinical trial. *JAMA Intern Med*. 2024;184(9):1106–1115.
22. Hill K, Boyer EW, Grundmann O, et al. De facto opioids: characterization of novel 7-hydroxymitragynine and mitragynine pseudoindoxyl product marketing. *Drug Alcohol Depend*. 2025;272:112701.

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